

$$\text{Prob} = \prod_{k=1}^5 \frac{1}{2\pi\sigma_{k,u}\sigma_{k,v}} \exp\left[-\frac{1}{2}\left(\frac{(u-\mu_{k,u})^2}{\sigma_{k,u}^2} + \frac{(v-\mu_{k,v})^2}{\sigma_{k,v}^2}\right)\right],$$

$$(w_1, \dots, w_5) = (0.22, 0.19, 0.21, 0.18, 0.20), \quad \sum_{k=1}^5 w_k = 1,$$

$$\mu_1 = (-1.75, 0.15), \quad \mu_2 = (-0.70, -1.55), \quad \mu_3 = (0.75, 1.45), \quad \mu_4 = (1.70, -0.55), \quad \mu_5 = (-0.10, 0.65).$$

